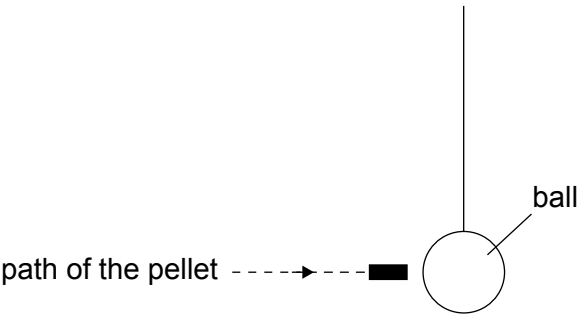


12. A stationary ball is hanging from a light string. A pellet from an air rifle is travelling horizontally and becomes embedded in the ball. The velocity of the pellet when it strikes the ball is 160 ms^{-1} .



The following data are given.

Mass of the ball = 250 g

Mass of the pellet = 2.0 g

- (a) Calculate the speed of the ball and the pellet immediately after the impact. [2]

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- (b) Suggest why the combined kinetic energy of the ball and the pellet after the impact is less than the initial kinetic energy of the pellet. [2]

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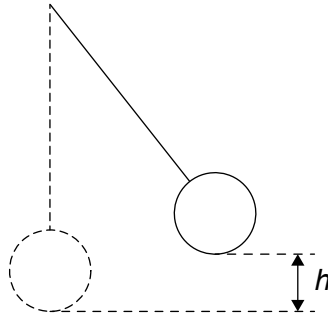
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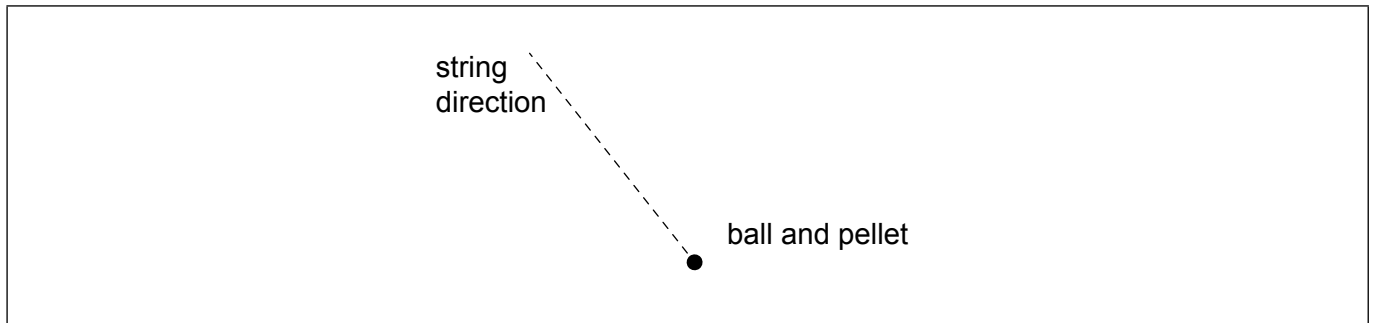
(Question 12 continued)

The ball with the embedded pellet rises to a maximum vertical height h .



- (c) Draw and label the free-body diagram for the ball at height h .

[2]



- (d) Determine h .

[2]

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